NameSayan

Roll NO.:2401010163

Section-B

Course:B.tech Cse

Core

Assignment 4 :

❖Aim

To design and develop a full-stack To-Do List web application using HTML, CSS, JavaScript, Python, and Flask, enabling users to add, edit, delete, mark complete, and manage tasks efficiently.

❖ Tasks (Flask Version)

• Set up a Flask project structure
• Create a Task model using a database (SQLite with SQLAlchemy)
• Design HTML templates using Jinja2
• Implement backend logic using Flask routes
• Create REST API endpoints (GET, POST, PUT, DELETE)
• Use JavaScript Fetch API for dynamic updates (no reload)
• Add task operations (create, update, delete, toggle)
• Implement filtering (All, Active, Completed)

❖ Technologies Used

• Frontend: HTML5, CSS3, JavaScript (ES6)
• Backend: Python 3
• Framework: Flask
• Database: SQLite (via SQLAlchemy)
• API Handling: Fetch API (AJAX)

❖ Code :

➤ **HTML CODE :**

❖ **base.html**

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>{% block title %}Flask ToDo App{% endblock %}</title>
   <link rel="stylesheet" href="{{ url_for('static', filename='css/style.css') }}">
   <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.4.0/css/all.min.css">
</head>
<body>
   <div class="container">
      <header>
         <h1><i class="fas fa-check-circle"></i> My ToDo List</h1>
         <p class="subtitle">Stay organized, stay productive</p>
      </header>

      {% block content %}{% endblock %}
   </div>

   <script src="{{ url_for('static', filename='js/main.js') }}"></script>
</body>
</html>
```

❖ **index.html**

```
<section class="add-todo-section">
  <form action="{{ url_for('add') }}" method="POST" class="add-form">
    <div class="form-group">
      <input
        type="text"
        name="title"
        placeholder="What needs to be done?"
        class="todo-input"
        required
        autofocus
    />
    </div>
    <div class="form-row">
      <div class="form-group flex-1">
        <input
          type="text"
          name="description"
          placeholder="Add a description (optional)"
          class="desc-input"
      />
      </div>
      <div class="form-group">
        <select name="priority" class="priority-select">
          <option value="low">🟢 Low</option>
          <option value="medium" selected>🟡 Medium</option>
          <option value="high">🔴 High</option>
        </select>
      </div>
```

```
            <button type="submit" class="btn btn-primary">
                <i class="fas fa-plus"></i> Add Task
            </button>
        </div>
    </form>
</section>

<!-- Stats Bar -->
<section class="stats-bar">
    <div class="stat-item">
        <span class="stat-number">{{ total }}</span>
        <span class="stat-label">Total</span>
    </div>
    <div class="stat-item active">
        <span class="stat-number">{{ active }}</span>
        <span class="stat-label">Active</span>
    </div>
    <div class="stat-item done">
        <span class="stat-number">{{ done }}</span>
        <span class="stat-label">Done</span>
    </div>
</section>

<!-- Filter Tabs -->
<section class="filter-tabs">
    <a href="{{ url_for('index', filter='all') }}"
       class="tab {% if filter_by == 'all' %}active{% endif %}">All</a>
    <a href="{{ url_for('index', filter='active') }}"
       class="tab {% if filter_by == 'active' %}active{% endif %}">Active</a>
```

```
    <a href="{{ url_for('index', filter='completed') }}"
       class="tab {% if filter_by == 'completed' %}active{% endif %}">Completed</a>


    {% if done > 0 %}
    <a href="{{ url_for('clear_completed') }}" class="tab tab-danger"
       onclick="return confirm('Delete all completed tasks?')">
        <i class="fas fa-trash"></i> Clear Completed
    </a>
    {% endif %}
</section>
<section class="todo-list">
    {% if todos %}
        {% for todo in todos %}
        <div class="todo-item {% if todo.completed %}completed{% endif %} priority-
{{ todo.priority }}">
            <div class="todo-left">
                <a href="{{ url_for('toggle', todo_id=todo.id) }}" class="toggle-btn"
                   title="{{ 'Mark as active' if todo.completed else 'Mark as done' }}">
                    <i class="fas {% if todo.completed %}fa-check-circle{% else %}fa-circle{%
endif %}"></i>
                </a>
                <div class="todo-content">
                    <span class="todo-title">{{ todo.title }}</span>
                    {% if todo.description %}
                    <span class="todo-desc">{{ todo.description }}</span>
                    {% endif %}
                    <span class="todo-meta">
                        <span class="badge priority-badge-
{{ todo.priority }}">{{ todo.priority }}</span>
                        <span class="todo-date">{{ todo.created_at.strftime('%b %d, %Y') }}</span>
```

```
                </span>
            </div>
        </div>
        <div class="todo-actions">
            <a href="{{ url_for('edit', todo_id=todo.id) }}" class="action-btn edit-btn"
title="Edit">
                <i class="fas fa-pen"></i>
            </a>
            <a href="{{ url_for('delete', todo_id=todo.id) }}" class="action-btn delete-btn"
title="Delete"
               onclick="return confirm('Delete this task?')">
                <i class="fas fa-trash"></i>
            </a>
        </div>
    </div>
    {% endfor %}
  {% else %}
    <div class="empty-state">
        <i class="fas fa-clipboard-list"></i>
        <p>No tasks found. Add one above!</p>
    </div>
  {% endif %}
</section>
{% endblock %}
```

❖ edit.html

```
<section class="edit-section">
  <div class="edit-card">
    <h2><i class="fas fa-pen"></i> Edit Task</h2>
```

```
<form action="{{ url_for('edit', todo_id=todo.id) }}" method="POST">
    <div class="form-group">
        <label for="title">Task Title *</label>
        <input
            type="text"
            id="title"
            name="title"
            value="{{ todo.title }}"
            class="todo-input"
            required
            autofocus
        />
    </div>

    <div class="form-group">
        <label for="description">Description</label>
        <textarea
            id="description"
            name="description"
            class="desc-textarea"
            placeholder="Add a description (optional)"
            rows="3"
        >{{ todo.description or '' }}</textarea>
    </div>

    <div class="form-group">
        <label for="priority">Priority</label>
        <select id="priority" name="priority" class="priority-select">
```

```
                <option value="low" {% if todo.priority == 'low' %}selected{% endif %}>
Low</option>
                <option value="medium" {% if todo.priority == 'medium' %}selected{%
endif %}> Medium</option>
                <option value="high" {% if todo.priority == 'high' %}selected{% endif %}>
High</option>
            </select>
        </div>

        <div class="form-actions">
            <button type="submit" class="btn btn-primary">
                <i class="fas fa-save"></i> Save Changes
            </button>
            <a href="{{ url_for('index') }}" class="btn btn-secondary">
                <i class="fas fa-times"></i> Cancel
            </a>
        </div>
    </form>
</div>
</section>
{% endblock %}
```

➤ **CSS CODE :**

```
:root {
   --primary:     #6c63ff;     --
   primary-dark:  #5a52d5;  -
   -danger:      #e74c3c;      --
   success:      #27ae60;      --
   warning:  #f39c12;  --low:
   #27ae60;
```

```
    --medium: #f39c12;
    --high: #e74c3c;
    --bg: #f0f2f5;
    --card: #ffffff;
    --text: #2d3436;
    --text-muted: #636e72;
    --border: #dfe6e9;
    --shadow: 0 2px 12px rgba(0,0,0,0.08);
    --radius: 12px;
}
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}
body {
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    background: var(--bg);
    color: var(--text);
    min-height: 100vh;
    padding: 20px;
}
.container {
    max-width: 700px;
    margin: 0 auto;
}
header {
    text-align: center;
    padding: 40px 0 30px;
```

```
}
header h1 {
    font-size: 2.2rem;
    color: var(--primary);
    margin-bottom: 6px;
}
header h1 i {
    margin-right: 10px;
}
.subtitle {
    color: var(--text-muted);
    font-size: 0.95rem;
}
.add-todo-section {
    background: var(--card);
    border-radius: var(--radius);
    padding: 24px;
    margin-bottom: 20px;
    box-shadow: var(--shadow);
}
.add-form .form-group {
    margin-bottom: 12px;
}
.form-row {
    display: flex;
    gap: 10px;
    align-items: center;
    flex-wrap: wrap;
}
```

```
.flex-1 { flex: 1; }
.todo-input, .desc-input, .priority-select, .desc-textarea {
    width: 100%;
    padding: 10px 14px;
    border: 2px solid var(--border);
    border-radius: 8px;
    font-size: 0.95rem;
    color: var(--text);
    transition: border-color 0.2s;
    background: #fafafa;
}
.todo-input {
    font-size: 1.05rem;
    padding: 12px 16px;
}
.todo-input:focus, .desc-input:focus, .priority-select:focus, .desc-textarea:focus {
    outline: none;
    border-color: var(--primary);
    background: #fff;
}
.priority-select {
    width: auto;
    cursor: pointer;
}
.btn {
    display: inline-flex;
    align-items: center;
    gap: 6px;
    padding: 10px 18px;
```

```
    border: none;
    border-radius: 8px;
    transition: all 0.2s;
    white-space: nowrap;
}
.btn-primary {
    background: var(--primary);
    color: white;
}
.btn-primary:hover {
    background: var(--primary-dark);
    transform: translateY(-1px);
}
.btn-secondary {
    background: var(--border);
    color: var(--text);
}
.btn-secondary:hover {
    background: #ccc;
}
.stats-bar {
    display: flex;
    gap: 12px;
    margin-bottom: 16px;
}

.stat-item {
flex: 1;
background: var(--card);
```

```
    border-radius: var(--radius);
    padding: 14px;
    text-align: center;
    box-shadow: var(--shadow);
}
.stat-number {
    display: block;
    font-size: 1.8rem;
    font-weight: 700;
    color: var(--text);
}
.stat-item.active .stat-number { color: var(--primary); }
.stat-item.done .stat-number { color: var(--success); }
.stat-label {
    font-size: 0.8rem;
    color: var(--text-muted);
    text-transform: uppercase;
    letter-spacing: 0.5px;
}
.filter-tabs {
    display: flex;
    gap: 8px;
    margin-bottom: 16px;
    flex-wrap: wrap;
}
.tab {
    padding: 8px 16px;
    border-radius: 20px;
    font-size: 0.88rem;
```

```
  font-weight:      500;      text-
  decoration: none;
  color:          var(--text-muted);
  background:          var(--card);
  border: 2px solid transparent;
  transition: all 0.2s;
}
.tab:hover { border-color: var(--primary); color: var(--primary); }
.tab.active { background: var(--primary); color: white; }
.tab-danger { color: var(--danger) !important; margin-left: auto; }
.tab-danger:hover { border-color: var(--danger); background: var(--danger); color:
white !important; }
.todo-list {
  display: flex;
  flex-direction: column;
  gap: 10px;
}

.todo-item {
  background: var(--card);
  border-radius: var(--radius);
  padding: 16px 18px;
  display: flex;
  align-items: center;
  justify-content: space-between;
  box-shadow: var(--shadow);
  border-left: 4px solid transparent;
  transition: all 0.2s;
}
.todo-item:hover { transform: translateY(-1px); box-shadow: 0 4px 18px rgba(0,0,0,0.12); }
```

```
.todo-item.priority-low { border-left-color: var(--low); }
.todo-item.completed { opacity: 0.6; }
.todo-item.completed .todo-title { text-decoration: line-through; color: var(--text-muted); }
.todo-left {
  display: flex;
  align-items: flex-start;
  gap: 14px;
  flex: 1;
}
.toggle-btn {
  font-size: 1.4rem;
  text-decoration: none;
  color: var(--border);
  transition: color 0.2s;
  padding-top: 2px;
  flex-shrink: 0;
}
.completed .toggle-btn { color: var(--success); }
.toggle-btn:hover { color: var(--primary); }
.todo-content {
  display: flex;
  flex-direction: column;
  gap: 4px;
}
.todo-title {
  font-size: 1rem;
  font-weight: 500;
}
.todo-desc {
```

```
    font-size: 0.85rem;
    color: var(--text-muted);
}
.todo-meta {
    display: flex;
    align-items: center;
    gap: 10px;
    margin-top: 2px;
}
.badge {
    font-size: 0.72rem;
    padding: 2px 8px;
    border-radius: 10px;
    font-weight: 600;
    text-transform: uppercase;
    letter-spacing: 0.4px;
}
.priority-badge-low { background: #d5f5e3; color: var(--low); }
.priority-badge-medium { background: #fef9e7; color: var(--medium); }
.priority-badge-high { background: #fdecea; color: var(--high); }

.todo-date {
    font-size: 0.78rem;
    color: var(--text-muted);
}
.todo-actions {
display: flex;
gap: 6px;
opacity: 0;
```

```
    transition: opacity 0.2s;
}

.todo-item:hover .todo-actions { opacity: 1; }
.action-btn {
    width: 32px;
    height: 32px;
    display: flex;
    align-items: center;
    justify-content: center;
    border-radius: 6px;
    text-decoration: none;
    font-size: 0.85rem;
    transition: all 0.2s;


}
.empty-state i {
    font-size: 3rem;
    margin-bottom: 16px;
    display: block;
    opacity: 0.4;
}
.edit-section {
    padding: 20px 0;
}
.edit-card {
    background: var(--card);
    border-radius: var(--radius);
    padding: 32px;
```

```
    box-shadow: var(--shadow);
}
.edit-card h2 {
    font-size: 1.4rem;
    color: var(--primary);
    margin-bottom: 24px;
}
.edit-card label {
display: block;
font-size: 0.88rem;
font-weight: 600;
margin-bottom: 6px;
color: var(--text-muted);
text-transform: uppercase;
letter-spacing: 0.4px;
}
.edit-card .form-group {
    margin-bottom: 18px;
}
.desc-textarea {
    resize: vertical;
    font-family: inherit;
}
.form-actions {
display: flex;
gap: 10px;
margin-top: 8px;
}
@media (max-width: 500px) {
```

```
    body { padding: 12px; }
    header h1 { font-size: 1.6rem; }
    .stats-bar { gap: 8px; }
    .form-row { flex-direction: column; }
    .form-row .btn { width: 100%; justify-content: center; }
    .todo-actions { opacity: 1; }
}
```

➤ **JS CODE:**

```
document.addEventListener('DOMContentLoaded', function () {
    const flashMessages = document.querySelectorAll('.flash-message');
    flashMessages.forEach(msg => {
        setTimeout(() => {
            msg.style.opacity = '0';
            setTimeout(() => msg.remove(), 500);
        }, 3000);
    });
});
```

➤ **APP.PY CODE:**

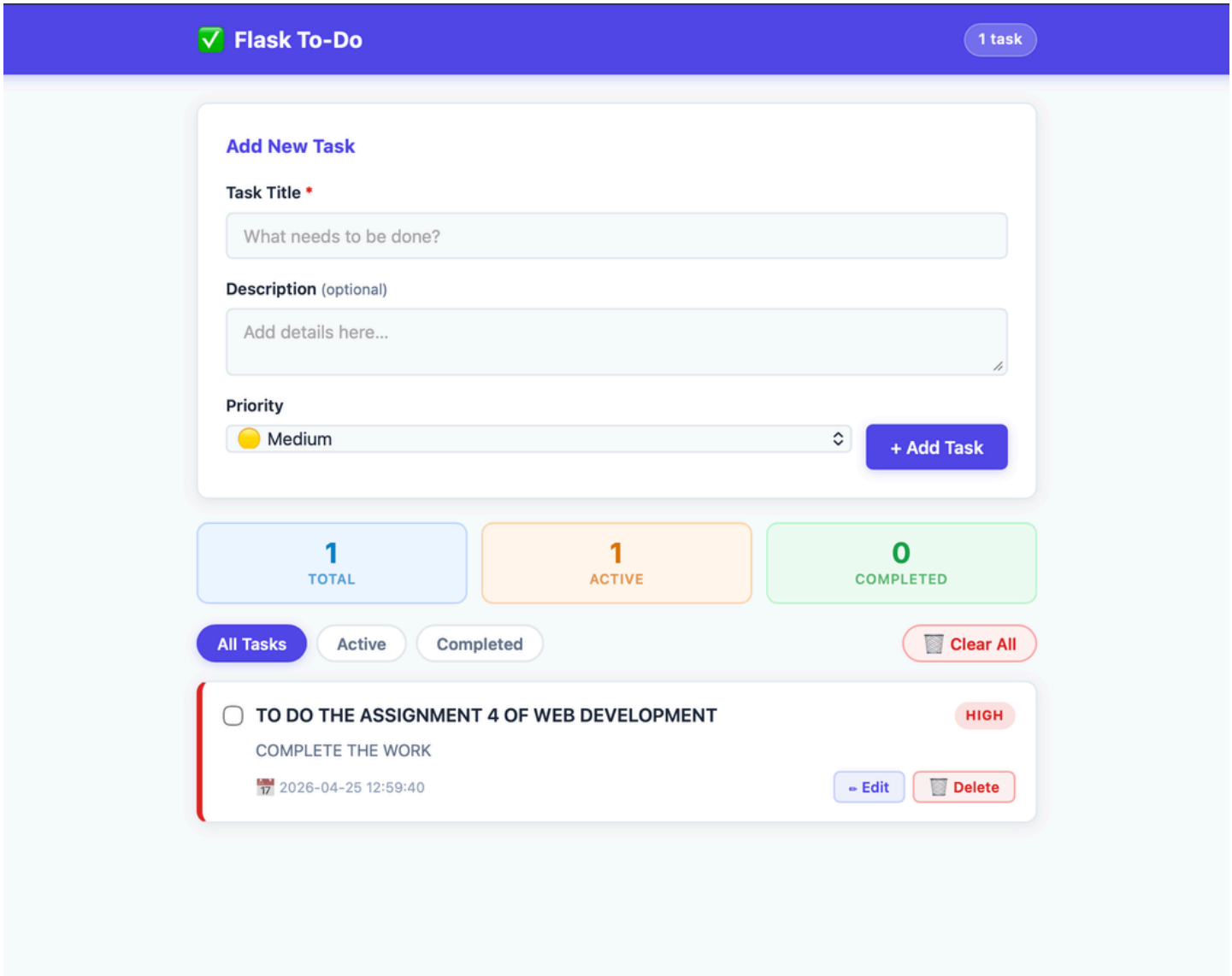
```
from flask import Flask, render_template, request, redirect, url_for
from flask_sqlalchemy import SQLAlchemy
from datetime import datetime
app = Flask(__name__)
app.config['SQLALCHEMY_DATABASE_URI'] = 'sqlite:///todos.db'
app.config['SQLALCHEMY_TRACK_MODIFICATIONS'] = False
```

```
db = SQLAlchemy(app)

class Todo(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    title = db.Column(db.String(200), nullable=False)
    description = db.Column(db.Text, nullable=True)
    completed = db.Column(db.Boolean, default=False)
    priority = db.Column(db.String(10), default='medium') # low, medium, high
    created_at = db.Column(db.DateTime, default=datetime.utcnow)
    def __repr__(self):
        return f'<Todo {self.id}: {self.title}>'
@app.route('/')
def index():
    filter_by = request.args.get('filter', 'all')
    if filter_by == 'active':
        todos = Todo.query.filter_by(completed=False).order_by(Todo.created_at.desc()).all()
    elif filter_by == 'completed':
        todos = Todo.query.filter_by(completed=True).order_by(Todo.created_at.desc()).all()
    else:
        todos = Todo.query.order_by(Todo.created_at.desc()).all()

    total = Todo.query.count()
    active = Todo.query.filter_by(completed=False).count()
    done = Todo.query.filter_by(completed=True).count()

    return render_template('index.html', todos=todos, filter_by=filter_by,
                           total=total, active=active, done=done)
@app.route('/add', methods=['POST'])
def add():
```

```
    title = request.form.get('title', '').strip()
    description = request.form.get('description', '').strip()
    priority = request.form.get('priority', 'medium')

    if title:
        new_todo = Todo(title=title, description=description, priority=priority)
        db.session.add(new_todo)
        db.session.commit()
    return redirect(url_for('index'))
@app.route('/toggle/<int:todo_id>')
def toggle(todo_id):
    todo = Todo.query.get_or_404(todo_id)
    todo.completed = not todo.completed
    db.session.commit()
    return redirect(request.referrer or url_for('index'))

@app.route('/edit/<int:todo_id>', methods=['GET', 'POST'])
def edit(todo_id):
    todo = Todo.query.get_or_404(todo_id)
    if request.method == 'POST':
        title = request.form.get('title', '').strip()
        if title:
            todo.title = title
            todo.description = request.form.get('description', '').strip()
            todo.priority = request.form.get('priority', 'medium')
            db.session.commit()
        return redirect(url_for('index'))
    return render_template('edit.html', todo=todo)
```

```
@app.route('/delete/<int:todo_id>')
def delete(todo_id):
    todo = Todo.query.get_or_404(todo_id)
    db.session.delete(todo)
    db.session.commit()
    return redirect(request.referrer or url_for('index'))

@app.route('/clear_completed')
def clear_completed():
    Todo.query.filter_by(completed=True).delete()
    db.session.commit()
    return redirect(url_for('index'))

if __name__ == '__main__':
    with app.app_context():
        db.create_all()
    app.run(debug=True)
```

➤ **OUTPUT :**


Flask To-Do
1 task
Add New Task
Task Title *
What needs to be done?
Description (optional)
Add details here...
Priority
Medium
+ Add Task
1
TOTAL
1
ACTIVE
0
COMPLETED
All Tasks
Active
Completed
Clear All
TO DO THE ASSIGNMENT 4 OF WEB DEVELOPMENT
HIGH
COMPLETE THE WORK
2026-04-25 12:59:40
Edit
Delete